MOBILE DRILL BRAIN

Calibration Technician Training Module 009

Training Progress, Supervised Runs, and Solo Service Approval

Field	Value
Document type	Workspace 006-B technician training module
Module	009 - Training Progress, Supervised Runs, and Solo Service Approval
Version	v0.1
Prepared	July 2026
Source basis	Calibration Service Module Extraction Worksheet v0.1; MDB Master Repository Unabridged; Modules 001-008 SME-approved deliverables
Training lane	Calibration Technician Training
Status	Deliverable draft for review; training-progress/approval framework, not a live scheduling authority

Locked module purpose
Teach how a new calibration technician moves from module training into observed field work, supervised data collection, report practice, review, and approval for solo service appointments. This module does not create live scheduling authority by itself.

1. Source basis and training boundary

This module is based on the Calibration Service Module Extraction Worksheet, the active Workspace 006 training-progress structure, and the completed Modules 001-008 in this calibration technician training lane.

The training record concept is already part of the calibration source structure: technician training should track written tests, observed calibrations, supervised data collection, report-building practice, trainer review, and approval before solo work.

Training boundary: this module defines progression and approval evidence. It does not make MDB the live scheduling authority, the official report archive, or a replacement for senior calibration tech judgment.

2. Learning outcomes

- Understand what training records must be completed before solo approval.
- Explain why module completion alone does not equal field readiness.
- Understand observed/work-with appointments, supervised lead appointments, and correction cycles.
- Understand the report-building observation and historical report run-through requirements.
- Recognize what kinds of mistakes delay approval for solo service appointments.
- Understand why high-priority job eligibility remains a senior calibration tech judgment call.
- Know how the Training Progress Tracker supports signoff.

3. Training progression overview

Step	Stage	Pass / evidence standard
1	Module completion	Modules 001-008 completed with knowledge checks.
2	Observed / work-with appointments	Minimum 5 observed/work-with calibration appointments; more if comprehension requires it.

3	Comprehension check	Senior calibration tech confirms trainee understands field sequence, data quality, documentation, and communication boundaries.
4	Supervised lead appointments	Minimum 5 appointments where trainee performs the work while senior calibration tech observes.
5	Report-building observation	Trainee observes senior calibration tech build at least 5 reports.
6	Historical report run-throughs	Minimum 4 historical run-throughs; trainee results are compared against original reports.
7	Trainee-built reports from own data	Senior calibration tech observes trainee building reports from trainee's own collected data; minimum remains judgment-based.
8	Troubleshooting scenario review	Trainee passes practical scenarios for failed pulse check, refusal, rapid advancement, damaged equipment, customer pressure, and report issues.
9	Approval for solo service appointments	Senior calibration tech signs off when evidence supports safe, repeatable, reviewable work.
10	High-priority eligibility	Separate open-ended senior calibration tech determination after proven performance over time.

Core rule
A technician is not approved for solo service appointments just because the modules were completed. Approval comes from repeated clean field performance, usable documentation, sound judgment, and senior calibration tech signoff.

4. Module completion and knowledge checks

The trainee should complete Modules 001-008 before the approval pathway is considered complete.

- 001 - Role, Purpose, and Workflow Overview
- 002 - SPT Analyzer Kit, Instrumented Rods, and Job Readiness
- 003 - SPT Fundamentals, N-Value, and N60
- 004 - Field Setup and SPT Analyzer Software Workflow
- 005 - LP, LE, Depth Intervals, and Field Measurement Accuracy
- 006 - Data Quality, Refusal, Sensor Issues, and Troubleshooting Boundaries
- 007 - Report Handoff, Field Closeout, and Customer Relationship Notes
- 008 - Customer Communication and Do-Not-Promise Rules

Recommended pass standard: 85% or higher on module quizzes. Any missed critical safety, data-quality, or authority-boundary item requires trainer review even if the overall score is passing.

- Instrumented rod entering hollow stem auger / borehole risk
- Strain gauges not field-replaceable
- Failed pulse check troubleshooting and escalation
- Refusal definition
- LP, LE, and depth interval accuracy
- 6-inch blow-count documentation
- Final report handling boundary
- Do-not-promise rules

5. Observed / work-with appointments

The trainee must complete a minimum of 5 observed/work-with calibration appointments before progressing into supervised lead work. The minimum may increase depending on comprehension, confidence, attention to detail, and field judgment.

Observed/work-with appointments are not passive ride-alongs. The trainee should watch the full workflow, ask questions, and explain back what they observed.

- Customer/driller check-in
- Analyzer kit setup and equipment protection
- Software setup and data-file context
- Depth interval, LP, and LE handling
- Sensor tab and pulse check

- First 6 blows observation
- 6-inch blow-count recording
- Refusal and rapid advancement response
- End-test/file closeout
- Calibration log completion
- Report handoff
- Customer relationship closeout

Training rule
Observed appointments should show the trainee what good work looks like before they are expected to lead the workflow.

6. Comprehension check before supervised lead work

After the observed/work-with stage, the senior calibration tech should perform a comprehension check before allowing the trainee to lead work under supervision.

Comprehension area	Trainee should be able to explain
Field sequence	How the job moves from arrival to setup, collection, closeout, and handoff.
Software path	CollectWire, project/setup fields, Sensor tab, pulse check, pause/play, End Test.
Measurement accuracy	Depth interval, LP, LE, sample interval, and 6-inch blow-count documentation.
Data-quality issues	Failed pulse check, refusal, rapid advancement, bad data, reruns, and escalation.
Equipment protection	Instrumented rod protection, accelerometer vs. strain gauge damage, and cable/connector awareness.
Communication	Do-not-promise rules, report delivery boundary, and when to route questions upward.

7. Supervised lead appointments

After the comprehension check, the trainee must complete a minimum of 5 supervised lead appointments. In this stage, the trainee does the work while the senior calibration tech observes and corrects as needed.

The goal is not simply completing five jobs. The goal is proving repeatable habits.

Evaluation area	What the senior calibration tech watches for
Setup discipline	Kit readiness, cable handling, sensor connection, rod/anvil awareness, and safe sequence control.
Software workflow	Correct fields, correct sequence, pulse check judgment, and end-test closeout.
Measurement accuracy	Correct depth interval, sample interval, LP, LE, and rod type.
Data collection	First 6 blows observation, 6-inch blow counts, refusal/rapid advancement awareness.
Documentation	Calibration log, notes, field obscurities, photos/video when needed, and handoff clarity.
Communication	Driller directions, manager/customer communication, and no overpromising.
Judgment	When to rerun, document, continue, or escalate.

8. Field documentation review

The senior calibration tech should review each supervised job's calibration log and handoff materials.

- One calibration log per rig
- Correct customer, rig, borehole, and date information

- Correct depth intervals and sample intervals
- Correct LP and measured LE values
- 6-inch blow-count increments recorded
- Refusal documented when applicable
- Rapid advancement documented when applicable
- Reruns documented
- Sensor or equipment issues documented
- Photos/video included when useful or required
- Data files transferred or handed off correctly
- Notes clear enough for a report builder who was not on site

Training rule
If the report builder has to guess what happened, the field documentation was not good enough.

9. Report-building observation and practice

Report-building practice is part of the progression because the trainee must understand how field data becomes a usable report. This does not automatically make the trainee a qualified report handler.

Stage	Minimum / requirement	Purpose
Observed report builds	Trainee observes senior calibration tech build at least 5 reports.	Show how field logs, analyzer data, notes, and report standards come together.
Historical run-throughs	Minimum 4 historical run-throughs after observing report builds.	Trainee runs through old datasets and compares results against original reports.
Own-data report practice	Undetermined minimum; senior calibration tech watches trainee build reports from the trainee's own collected data.	Confirm that trainee understands how their own field habits affect report quality.

The senior calibration tech decides when report-building practice is sufficient. The decision should be based on accuracy, consistency, attention to field notes, and whether the trainee's results match established report standards.

10. Troubleshooting scenario review

The trainee must pass practical scenario review before approval for solo service appointments.

- Failed pulse check
- Backup accelerometer replacement
- Cord/connector suspected issue
- Strain gauge damage suspicion
- Refusal
- Rapid advancement
- Short stroke
- Abnormal hammer idle/RPM behavior
- Hydraulic leak or pressure concern
- Missed 6-inch blow counts
- Wrong depth interval
- Customer pressure to rush
- Customer asking for pass/fail result
- Urgent report request

Training rule
A solo service tech must know when to keep working, when to rerun, and when to stop and escalate.

11. Trainer notes and correction cycles

Training should include written notes from the senior calibration tech.

- Date
- Job/customer reference when appropriate
- Trainer and trainee

- What was observed
- What was done correctly
- What needs correction
- Whether the correction was retested
- Whether the issue is resolved
- Whether trainee can proceed to the next stage

Correction cycle: observe issue → explain correction → trainee repeats task → senior calibration tech verifies improvement → note whether resolved.

Training rule
A mistake does not automatically fail a trainee. Repeating the same mistake without correction delays approval.

12. Approval standard

Formal approval wording:

Approved for solo service appointments
This means the technician can safely collect usable calibration data on standard service appointments and knows when to escalate. It does not mean the technician is automatically eligible for high-priority work or independent report authority.

Approval should be based on four areas:

Area	Approval evidence
Technical setup	Kit readiness, software workflow, LP/LE/depth interval accuracy, and instrumented rod protection.
Data quality	Clean pulse checks, valid collection habits, refusal/rapid advancement handling, and rerun/escalation judgment.
Documentation / handoff	Complete calibration logs, data transfer, notes, and report-builder-ready field records.
Judgment / communication	Professional communication, no overpromising, customer pressure handling, and correct escalation.

13. High-priority job eligibility

High-priority job eligibility is separate from approval for solo service appointments.

High-priority eligibility remains open-ended and is determined by senior calibration tech judgment. It should be based on logged hours, data quality, report quality, judgment, customer communication, and proven performance over time.

High-priority jobs should stay with senior techs until a newer technician has collected enough data over time to prove they can perform at a high standard.

Training rule
Solo service approval is a step, not the finish line.

14. Post-approval monitoring

Early solo appointments should still be reviewed.

- Data file quality
- Calibration log completeness
- Field notes
- LP/LE consistency
- Depth interval and sample interval consistency
- Issue handling
- Customer communication notes
- Whether the report builder needed clarification

The first solo appointments are still part of the training curve.

15. Common reasons approval is delayed

Issue	Why approval may be delayed
Guessing LE	LE must be measured; guessing creates bad field context.
Copying LP/LE from prior interval	Each data set needs its own correct depth context.
Missing 6-inch blow counts	Software total does not replace report-needed field breakdown.
Pressing forward after failed pulse check	Bad sensor path can make the data unusable.
Instrumented rod risk	Equipment damage risk is too high.
Undocumented refusal	Refusal data cannot be treated as normal clean calibration data.
Incomplete handoff	Report builder may need clarification or may not be able to complete report cleanly.
Overpromising	Customer trust and internal authority boundaries are at risk.
Not escalating damage	Equipment damage and unresolved issues require senior/office involvement.
Repeating same mistake	Uncorrected habits show the trainee is not ready for solo service appointments.

16. Training scenario

Scenario

A trainee has completed Modules 001-008 and passed the quizzes. On a supervised appointment, they set up the analyzer correctly but forget to record 6-inch blow counts on paper for one interval. The software still records total blows. The trainee says, "The software has the total, so we're probably fine."

Correct trainer response should include:

- Module completion alone does not equal approval for solo service appointments.
- Software total does not replace 6-inch blow-count documentation.
- The missing record affects report-builder clarity.
- The issue should be documented.
- If caught on site and another valid interval can be collected, best practice is to collect another proper data set.
- The trainee needs correction and retest before approval.

17. Knowledge check

#	Question	Answer key
1	Does completing Modules 001-008 automatically approve a trainee for solo service appointments?	No. Field performance and senior calibration tech approval are required.
2	What is the minimum observed/work-with appointment count?	Minimum 5, possibly more depending on comprehension.
3	What happens after the comprehension check?	Minimum 5 supervised lead appointments where the trainee does the work while the senior calibration tech observes.
4	How many report builds must the trainee observe before historical run-throughs?	Minimum 5 report builds.
5	How many historical report run-throughs are required?	Minimum 4.
6	What is the formal approval wording?	Approved for solo service appointments.
7	Is high-priority job eligibility automatic after solo approval?	No. It remains a senior calibration tech judgment call.
8	Name two items that should delay approval.	Examples: guessed LE, missed blow counts, failed pulse check handled wrong, no escalation after damage, overpromising, incomplete handoff.
9	What is the standard for approval?	Safe, repeatable, reviewable work.
10	Who signs off the trainee's progression?	Senior calibration tech.

18. Practical signoff checklist

Signoff item	Pass standard	Senior calibration tech initials/date
Module completion	Trainee completed required modules and quizzes.	
Observed/work-with appointments	Minimum 5 completed; trainee can explain full workflow.	
Comprehension check	Trainee explains sequence, data quality, documentation, and communication boundaries.	
Supervised lead appointments	Minimum 5 completed with trainee doing the work under observation.	
LP/LE and depth accuracy	Trainee measures/enters values correctly.	
Blow-count documentation	Trainee records 6-inch increments correctly.	
Data-quality judgment	Trainee recognizes bad data, refusal, rapid advancement, and failed pulse checks.	
Equipment protection	Trainee protects instrumented rod and handles damage correctly.	
Report-building observation	Trainee observed at least 5 report builds.	
Historical report run-throughs	Minimum 4 completed and compared to original reports.	
Own-data report practice	Senior calibration tech observed trainee report practice on own collected data.	
Customer communication	Trainee avoids promises and communicates professionally.	
Scenario pass	Trainee passes troubleshooting and communication scenarios.	
Solo service approval	Senior calibration tech approves trainee for solo service appointments.	

Appendix A. Training Progress Tracker

Use this tracker as a blank record template. A working software version may later store the same information inside MDB/MDP training records.

A1. Trainee record

Field	Entry
Trainee name	
Training start date	
Senior calibration tech	
Current status	
Notes	

A2. Module completion and quizzes

Module	Completion date	Quiz score	Critical items reviewed?
Module 001			
Module 002			
Module 003			
Module 004			
Module 005			
Module 006			
Module 007			
Module 008			

A3. Observed / work-with appointments

#	Date	Customer/job	Observed focus	Senior tech initials
1				
2				
3				
4				
5				

A4. Supervised lead appointments

#	Date	Customer/job	Trainee performed full workflow?	Senior tech initials
1				
2				
3				
4				
5				

A5. Report-building progression

Progression item	Minimum	Completion evidence	Senior tech initials/date
Observed report builds	5		
Historical report run-throughs	4		
Trainee-built reports from own collected data	Senior tech judgment		

A6. Correction items

#	Date	Issue	Correction given	Retested?	Closed?
1					
2					
3					
4					
5					

A7. Approval record

Approval item	Decision / notes	Senior calibration tech initials/date
Approved for solo service appointments		
High-priority eligibility reviewed		
High-priority eligible?		
Next review date		

Revision history

Version	Status	Change
v0.1	Deliverable draft	Created Module 009 with SME-approved progression counts, solo service approval wording, high-priority eligibility boundary, senior calibration tech authority, and progress tracker appendix.